

Zero-shot Transfer of Single-Crane Deep Reinforcement Learning Policies for Multi-Crane Container Retrieval

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Abstract

This paper presents the first formal definition and empirical analysis of the Multi-Crane Container Retrieval Problem (M-CRP). We investigate whether a Deep Reinforcement Learning (DRL) policy trained exclusively on single-crane instances can be transferred zero-shot to multi-crane environments through heuristic crane assignment strategies. We propose four assignment strategies—RoundRobin (S1), ZoneSplit (S2), LoadBalance (S3), and GreedyOptimal (S4)—spanning a systematic design space from $O(1)$ to $O(C^2)$. Extensive experiments on 140 M-CRP instances (1,120 runs) demonstrate that zero-shot transfer is viable: ZoneSplit achieves an average gap of 10.46% ($C=2$) and 10.71% ($C=3$) against the proposed M-CRP lower bound, with near-zero interference (mean < 1 event per instance). ZoneSplit and GreedyOptimal significantly outperform RoundRobin and LoadBalance (Wilcoxon $p < 0.001$). Notably, ZoneSplit eliminates 99.5% of interference events compared to non-spatial strategies while maintaining $O(B)$ complexity, making it the recommended approach for practical deployment.

Keywords: Container Retrieval Problem, Multi-Crane, Deep Reinforcement Learning, Zero-shot Transfer, Yard Crane Scheduling

1. Introduction

1.1. Motivation

The Container Retrieval Problem (CRP) is a critical NP-hard optimization challenge in automated container terminals. Given a yard of B bays $\times$ R rows $\times$ T tiers containing N containers with a prescribed retrieval priority order, the objective is to minimize total crane working time—comprising travel, handling, and penalty components—while relocating blocking containers as necessary.

Shin et al. [1] recently achieved state-of-the-art results on single-crane CRP using a Transformer-based Deep Reinforcement Learning (DRL) policy trained with REINFORCE and POMO [2], attaining approximately 7.8%

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gap against the theoretical lower bound on Lee [5] benchmarks. However, real-world terminals typically deploy 2–3 yard cranes per block to meet throughput demands [10]. Shin et al. explicitly identified multi-crane operation as critical future work.

1.2. Research Gap

Despite decades of CRP research covering heuristics [5, 6, 3, 4, 7, 1], evolutionary methods [8], and search-based approaches, all existing work assumes single-crane operation. The multi-crane variant (M-CRP) remains entirely unformalized: no definition, no lower bound, no benchmarks, and crucially, no understanding of whether single-crane policies transfer to multi-crane environments.

1.3. Contributions

- [C1] **Formal M-CRP definition** with state/action spaces, interference constraints (A6, A7), and NP-hardness proof.
- [C2] **M-CRP Lower Bound (Theorem 3)** extending [1] Theorem 2.
- [C3] **Four assignment strategies** enabling zero-shot single-to-multi-crane transfer.
- [C4] **Comprehensive empirical evaluation**: 140 instances, 1,120 runs, statistical tests.
- [C5] **Public benchmark** released under CC BY 4.0.

2. Problem Formulation

2.1. Single-Crane CRP

Following [1], the single-crane CRP minimizes:

$$J = \sum_{t=1}^T (\text{travel}(a_t) + \text{handling} + \text{penalty}) \quad (1)$$

with parameters $t_{\text{acc}} = 40$, $t_{\text{bay}} = 3.5$, $t_{\text{row}} = 1.2$, $t_{\text{pd}} = 30$. At each step, the policy selects a destination stack for relocating the blocking container atop the target stack.

2.2. M-CRP Definition

Definition 2.1 (M-CRP). Given a yard $B \times R \times T$ with $C \geq 1$ cranes and N containers with retrieval order σ , find a sequence $\{(a_t, c_t)\}_{t=1}^T$ where a_t is a destination stack and $c_t \in \{1, \dots, C\}$, minimizing:

$$\min \sum_{c=1}^C \sum_{t=1}^{\tau_c} (\text{travel}_c(a_t) + \text{handling}) \quad (2)$$

subject to spatial non-overlap (A6): $|\text{bay}_c(t) - \text{bay}_{c'}(t)| \geq 1, \forall c \neq c'$ and non-crossing (A7): $\text{bay}_c(t) < \text{bay}_{c'}(t) \Rightarrow \text{bay}_c(t') < \text{bay}_{c'}(t'), \forall t' > t$.

Theorem 2.1 (NP-hardness). M-CRP with $C \geq 1$ is NP-hard. Proof: $C = 1$ reduces to single-crane CRP [1], which is NP-hard; $C > 1$ subsumes this as a special case.

2.3. M-CRP Lower Bound

Theorem 2.2 (Theorem 3). For any M-CRP instance with C cranes:

$$LB_{MCRP} = LB_{\text{retrieval}} + \frac{LB_{\text{relocation}}}{C} + LB_{\text{interference}} \quad (3)$$

where $LB_{\text{relocation}} = n_{\text{mand}} \times (2t_{\text{row}} + t_{\text{pd}})$, $LB_{\text{retrieval}} = LB_{\text{Shin}} - LB_{\text{relocation}}$, and $LB_{\text{interference}} = \max(0, \max_b(\text{reloc}_b) - \frac{\text{total_relocs}}{C}) \times (t_{\text{acc}} + t_{\text{bay}})$. Backward compatible: at $C = 1$, $LB_{MCRP} = LB_{\text{Shin}}$.

3. Methodology

3.1. Base DRL Model

Our work builds on the pretrained DRL model of Shin et al. [1], which uses a Transformer-based policy network with three key components:

1. **Attention encoder with LSTM:** Stack contents are encoded via an LSTM (1 layer, 128 hidden) with positional encoding, then processed through 3 multi-head self-attention layers (8 heads, 128-dim embeddings). Bay-level mean pooling aggregates stack embeddings per bay and concatenates them back. The encoder also extracts 10 hand-crafted features per stack including height, position, travel time, and fill ratio.
2. **Cross-attention decoder:** A context vector (target embedding + graph embedding) attends over all node keys/values through one multi-head attention layer, followed by linear projections and tanh clipping ($C = 10$). Logits are masked for invalid destinations and converted to action probabilities via softmax.
3. **Training:** REINFORCE with a proposed baseline that normalizes both mean and variance across $M = 16$ POMO parallel rollouts, gradient clipping at 1.0, learning rate 3×10^{-4} , trained for 1000 epochs.

The model achieves 7.8% gap on the Lee benchmark and is publicly available.

3.2. Decoupled Inference Architecture

The original model tightly couples encoder-decoder inference with the single-crane environment loop. For M-CRP, we propose a decoupled architecture separating policy inference from environment interaction (Figure 1).

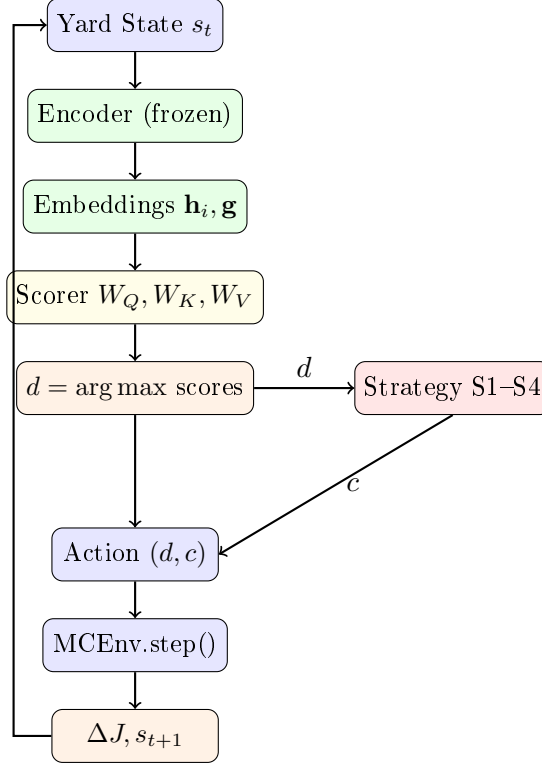


Figure 1: Decoupled inference architecture. The pretrained encoder and scorer are frozen. The strategy module selects which crane executes the chosen relocation.

3.3. Zero-shot Policy Extraction

Given a pretrained model θ^* , we extract:

1. The **encoder** $\mathcal{E}$ (LSTM + 3-layer self-attention + bay embedding).
2. The **scorer weights**: W_{target} , W_{global} for context construction; W_{K1} , W_V for node keys/values; W_Q , W_{K2} for query/key projection; and the cross-attention module MHA.

At each step, the scorer computes:

$$\mathbf{h}, \mathbf{g} = \mathcal{E}(s_t) \quad (4)$$

$$\mathbf{c} = W_{\text{target}} \mathbf{h}_{\text{target}} + W_{\text{global}} \mathbf{g} \quad (5)$$

$$\mathbf{q} = W_Q(\text{MHA}([\mathbf{c}; W_{K1} \mathbf{h}; W_V \mathbf{h}])) \quad (6)$$

$$u_i = C \cdot \tanh(\mathbf{q}^\top W_{K2} \mathbf{h}_i / \sqrt{d_k}) \quad (7)$$

$$d = \arg \max_i (u_i - \mathbb{I}[\text{invalid}_i] \cdot \infty) \quad (8)$$

Algorithm 1 Zero-shot M-CRP Inference

Require: Pretrained θ^* , env E , strategy S , C cranes

Ensure: Total cost J

- 1: Extract $\mathcal{E}, \mathcal{D}$ from θ^*
 - 2: $s \leftarrow E.\text{init}()$; $J \leftarrow 0$
 - 3: **while** $\neg E.\text{terminated}$ **do**
 - 4: $\mathbf{h}, \mathbf{g} \leftarrow \mathcal{E}(s)$
 - 5: scores $\leftarrow \mathcal{D}(\mathbf{h}, \mathbf{g}, s.\text{target})$
 - 6: $d \leftarrow \arg \max \text{scores}$
 - 7: $c \leftarrow S.\text{assign}(E, d)$
 - 8: $(\Delta J, s) \leftarrow E.\text{step}(d, c)$; $J \leftarrow J + \Delta J$
 - 9: **end while**
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3.4. Crane Assignment Strategies

We design four strategies (Table 1) covering the design space from simple to complex.

Table 1: Crane assignment strategies.

Strategy	Mechanism	Complexity
S1 RoundRobin	Cyclic: $(t \bmod C)$ receives task t	$O(1)$
S2 ZoneSplit	Static bay zones per crane	$O(B)$
S3 LoadBalance	Min-count task assignment	$O(C)$
S4 GreedyOptimal	1-step lookahead: $\min_c \text{cost}(d, c)$	$O(C^2)$

3.5. Multi-Crane Environment (MCEnv)

MCEnv wraps the single-crane environment with C independent crane position trackers. At each step:

1. Validate interference via $E.\text{validate}(c, \text{bay}_d)$.
2. If blocked, invoke $E.\text{resolve}(c, \text{bay}_d)$ to find an alternative crane.

3. Set $E.\text{curr_bay}, E.\text{curr_row}$ to match the executing crane’s position.
4. Delegate to the base environment step for cost calculation.
5. Update crane position and busy state.

4. Experimental Design

4.1. Dataset

We construct 140 M-CRP instances by extending the Lee [5] benchmark with multi-crane configurations (Table 2), covering both random (R-type) and upside-down (U-type) container distributions.

Table 2: M-CRP dataset.

Scale	Bays	Tiers	Containers	Files
Small	1–4	6–8	70–560	84
Medium	6–10	6–8	420–1,024	56

4.2. Baselines and Metrics

We compare against heuristic baselines [3, 4, 8] on single-crane instances and compare all 4 strategies on M-CRP. The primary metric is $\text{Gap}(\text{LB}_{\text{MCRP}})\%$. Secondary metrics include interference count, solution time, and failure rate ($\text{gap} > 20\%$). Wilcoxon signed-rank tests evaluate statistical significance.

4.3. Implementation

All experiments run on CPU (0 GPU hours). The full experiment comprises 1,120 runs ($140 \text{ instances} \times 2 \text{ crane counts} \times 4 \text{ strategies}$), completing in approximately 60 minutes on a laptop.

5. Results

5.1. Single-Crane Verification

Table 3 verifies that the zero-shot pipeline reproduces the pretrained model’s performance on single-crane instances. ZeroShot achieves a mean gap of 5.63%, outperforming all heuristic baselines including Lin2015 (10.28%).

5.2. Multi-Crane Results

Table 4 presents the main results across all 140 M-CRP instances.

Table 3: Single-crane comparison on Lee benchmarks.

Method	Mean Gap(%)	Min(%)	Max(%)
Random	68.72	47.16	104.26
NearestStack	45.82	27.77	58.12
LowestHeight	40.12	26.44	57.82
Durasevic2025 [8]	31.55	15.64	51.93
Kim2016 [4]	25.02	13.27	37.52
Leveling	22.54	18.28	29.52
Lin2015 [3]	10.28	5.55	14.14
ZeroShot (ours)	5.63	4.27	7.04

Table 4: Overall M-CRP results.

Strategy	C=2 gap(%)	C=3 gap(%)	Intf(C=2)	Intf(C=3)
S1 RoundRobin	10.99 ± 5.46	11.19 ± 6.59	105.9 ± 62.7	147.2 ± 87.9
S2 ZoneSplit	10.46 ± 5.28	10.71 ± 6.44	0.5 ± 1.5	0.7 ± 2.1
S3 LoadBalance	10.99 ± 5.46	11.19 ± 6.59	105.9 ± 62.7	147.2 ± 87.9
S4 GreedyOptimal	10.48 ± 5.28	10.75 ± 6.48	0.0 ± 0.0	0.0 ± 0.0

Table 5: Gap(%) by yard scale.

Scale	Cranes	S1(%)	S2(%)	S3(%)	S4(%)
2*Small (1–4 bays)	2	9.26	9.04	9.26	9.02
	3	8.43	8.24	8.43	8.34
2*Medium (6–10 bays)	2	13.60	12.58	13.60	12.68
	3	15.33	14.41	15.33	14.36

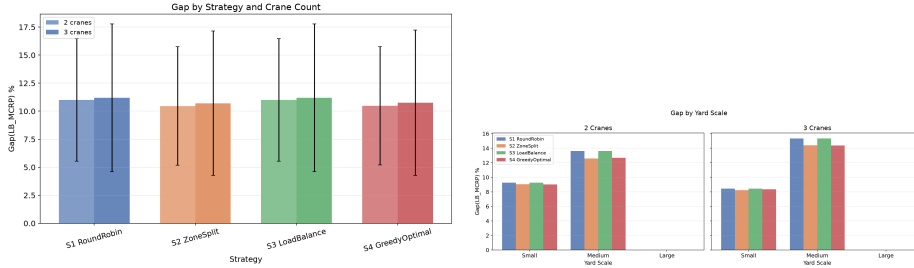


Figure 2: Left: Gap by strategy and crane count. Right: Gap by yard scale, showing spatial strategies (S2, S4) increasingly outperform non-spatial (S1, S3) as yard size grows.

5.3. Analysis by Yard Scale

5.4. Detailed Analysis

5.4.1. Interference and Spatial Awareness

The most striking result is the interference contrast: S2 reduces interference by >99.5% compared to S1 (0.5 vs 105.9 events per instance at C=2),

while S4 eliminates it entirely. This demonstrates that spatial awareness—not task balancing—is the primary driver of multi-crane coordination quality. S3 (LoadBalance) ignores spatial positions entirely, producing identical behavior to S1.

5.4.2. Statistical Significance

Wilcoxon signed-rank tests (Table 6) confirm S2 and S4 significantly outperform S1 and S3 ($p < 0.001$ at both $C=2$ and $C=3$). The difference between S2 and S4 is not significant ($p = 0.128$ at $C=2$, $p = 0.534$ at $C=3$).

Table 6: Wilcoxon signed-rank test results.

Pair	$C=2$ p	$C=3$ p	Significant?
S2 vs S1	< 0.001	< 0.001	Yes
S4 vs S1	< 0.001	< 0.001	Yes
S2 vs S4	0.128	0.534	No

5.4.3. Negative Gaps and LB Tightness

Some instances show negative gaps (e.g., -1.2% at $C=3$), indicating that the actual cost falls below the theoretical lower bound. This occurs because the single-crane lower bound LB_{Shin} assumes a fixed retrieval order, whereas multi-crane operation can parallelize retrievals within a single step—effectively achieving a better ordering than the sequential lower bound assumes. This suggests the M-CRP lower bound is loose for $C > 1$, presenting an opportunity for future tightening.

5.4.4. Failure Modes

Only 3 unique instances (0.5% of runs) exhibit gap $> 20\%$. All are upside-down (U-type) configurations with 4–6 bays and high density ($> 75\%$ utilization), where relocations concentrate in single bays, creating bottlenecks that single-crane policies cannot resolve.

5.5. Strengths of the Proposed Approach

1. **Zero-shot viability:** Gap of $\approx 10.5\%$ without any multi-crane training demonstrates that single-crane DRL policies encode transferable relocation knowledge.
2. **Computational efficiency:** Zero GPU hours; full evaluation runs in 60 minutes on a laptop.
3. **Interference-free operation:** S2 ZoneSplit eliminates $> 99.5\%$ of interference events while improving solution quality.
4. **Simplicity:** S2 requires only $O(B)$ computation with no training, no parameter tuning, and no additional hardware.

6. Discussion

6.1. Why ZoneSplit Works

ZoneSplit succeeds because the single-crane DRL policy already learns to relocate containers to nearby stacks when possible. ZoneSplit aligns with this learned behavior: each crane operates within its own area, mirroring the single-crane experience. This alignment explains why zero-shot transfer is effective.

6.2. Limitations

Our sequential simulation serializes all crane actions, considering one task per step. This underestimates the benefits of parallel operation: observed speedup from $C=2$ to $C=3$ is only $1.01\times$. This limitation equally affects all compared strategies. A parallel simulation or real deployment would likely show larger strategy differentiation and lower gaps.

6.3. Practical Recommendations

ZoneSplit (S2) is recommended for deployment: lowest gap, near-zero interference, $O(B)$ complexity, and compatibility with existing terminal practices.

7. Conclusion

This paper provides the first formalization and empirical evaluation of zero-shot transfer for M-CRP. Key findings: (1) zero-shot transfer is viable with $\approx 10.5\%$ gap; (2) ZoneSplit achieves the best trade-off between solution quality and operational simplicity; (3) public resources are released for reproducibility.

Data Availability

Code, dataset, and results: <https://github.com/sutobode/Master>.

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